

# Experimental Performance Analysis of Supervised Learning Algorithms using Friedman and Nemenyi Tests

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## I. INTRODUCTION

Evaluating the performance of supervised learning algorithms is critical for selecting the most appropriate model for a given task. This report presents the results of comparing three classifiers: Logistic Regression, Random Forest, and Support Vector Machine (SVM). The evaluation considers three performance measures: training time, accuracy, and F1-score. Statistical tests, specifically the Friedman and Nemenyi tests, are employed to determine whether these algorithms exhibit significant differences in performance.

## II. EXPERIMENTAL SETUP

### A. Data

The experiments utilized the Spambase dataset from the UCI Machine Learning Repository. The dataset contains numeric features extracted from email messages and a binary target variable indicating whether an email is spam or not.

### B. Classifiers

- *Logistic Regression (LR)*: Logistic Regression is a fundamental linear model for binary classification. It applies the sigmoid function to predict probabilities and uses maximum likelihood estimation for parameter fitting.
- *Random Forest (RF)*: Random Forest is an ensemble learning method that combines the predictions of multiple decision trees to improve accuracy and reduce overfitting. It operates by creating a 'forest' of trees, where each tree is trained on a bootstrap sample of the data and uses a subset of features for splitting.
- *Support Vector Machine (SVM)*: Support Vector Machine is a powerful classification algorithm that uses a radial basis function (RBF) kernel to map data into a higher-dimensional space. It finds the optimal hyperplane that maximizes the margin between the classes, making it effective for non-linear decision boundaries.

### C. Evaluation Measures

The following measures were used:

- 1) *Training Time*: Time taken to train the model.
- 2) *Accuracy*: Proportion of correctly classified instances.
- 3) *F1-Score*: Harmonic mean of precision and recall.

### D. Methodology

Stratified 10-fold cross-validation was used to evaluate each algorithm. For each fold, performance values and ranks were recorded. The experiments were conducted in a systematic manner:

1) *Dataset*: The Spambase dataset from the UCI Machine Learning Repository was utilized, containing numeric features extracted from email messages and a binary target variable indicating whether an email is spam or not.

2) *Training and Evaluation*: Each algorithm (Logistic Regression, Random Forest, and SVM) was trained on 90% of the data (9 folds) and tested on the remaining 10% (1 fold). This process was repeated for 10 stratified folds.

3) *Performance Recording*: For each fold, the training time, accuracy, and F1-score were recorded for all three algorithms. The ranks were assigned based on their relative performance.

4) *Statistical Analysis*: The Friedman test was performed to determine if there were statistically significant differences in performance. If significant, the Nemenyi test was conducted to identify which pairs of algorithms showed meaningful differences.

## III. STATISTICAL TESTS

### A. Friedman Test

The Friedman test is a non-parametric statistical test developed by Milton Friedman. Similar to the parametric repeated measures ANOVA, it is used to detect differences in treatments across multiple test attempts. The procedure involves ranking each row (or block) together, then considering the values of ranks by columns. The Friedman test evaluates whether the rankings of algorithms differ significantly across datasets.

$$\chi_F^2 = \frac{12N}{k(k+1)} \left( \sum_{j=1}^k \bar{R}_j^2 - \frac{k(k+1)^2}{4} \right)$$

Where:

N is the number of datasets/folds

K is the number of algorithms

$\bar{R}_j$  is the average rank of the j-th algorithm.

#### B. Nemenyi Test

The Nemenyi test is a post-hoc statistical test that can be used to determine which groups are significantly different from each other after an initial Friedman test has been performed and has rejected the null hypothesis that the performance of the comparisons on the groups of data is similar. The test makes pair-wise tests of performance.

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}}$$

Where:

$q_\alpha=2.343$  (critical value from the Studentized range distribution for  $k=3$  and  $\alpha=0.05$ )

K is the number of algorithms

N is the number of datasets/folds

### IV. RESULTS

#### A. Training Time

TABLE I. TRAINING TIME RESULTS

| Data set | Logistic Regression | Random Forest | SVM        |
|----------|---------------------|---------------|------------|
| 1        | 1.1349 (3)          | 0.3764 (2)    | 0.2295 (1) |
| 2        | 1.1886 (3)          | 0.3736 (2)    | 0.2556 (1) |
| 3        | 0.8780 (3)          | 0.3869 (2)    | 0.2654 (1) |
| 4        | 1.1104 (3)          | 0.4032 (2)    | 0.2266 (1) |
| 5        | 1.3505 (3)          | 0.4270 (2)    | 0.2836 (1) |
| 6        | 1.2718 (3)          | 0.3562 (1)    | 0.3665 (2) |
| 7        | 1.0872 (3)          | 0.3514 (2)    | 0.2507 (1) |
| 8        | 1.0501 (3)          | 0.3454 (2)    | 0.2531 (1) |
| 9        | 0.7080 (3)          | 0.3621 (2)    | 0.2486 (1) |
| 10       | 1.0471 (3)          | 0.3641 (2)    | 0.3522 (1) |
| avg rank | 3.0                 | 1.9           | 1.1        |

The Friedman test revealed a significant difference in training times ( $\chi_F^2 = 18.20 > 7.8$ ,  $p < 0.05$ ), confirming that the algorithms differ significantly in terms of training time. The Nemenyi test identified a significant difference between Logistic Regression vs. SVM (diff = 2.0 > CD = 1.0478). However, no significant differences were observed between Logistic Regression and Random Forest (diff = 1.0 < CD) and Random Forest and SVM (diff = 1.0 < CD).

#### B. Accuracy

TABLE II. ACCURACY RESULTS

| Data set | Logistic Regression | Random Forest | SVM        |
|----------|---------------------|---------------|------------|
| 1        | 0.9132 (2)          | 0.9523 (1)    | 0.7158 (3) |
| 2        | 0.9239 (2)          | 0.9565 (1)    | 0.7196 (3) |
| 3        | 0.9326 (2)          | 0.9609 (1)    | 0.7348 (3) |
| 4        | 0.9370 (2)          | 0.9674 (1)    | 0.7109 (3) |
| 5        | 0.9130 (2)          | 0.9457 (1)    | 0.7196 (3) |
| 6        | 0.9391 (2)          | 0.9565 (1)    | 0.7391 (3) |
| 7        | 0.9348 (2)          | 0.9587 (1)    | 0.6935 (3) |
| 8        | 0.9304 (2)          | 0.9565 (1)    | 0.7065 (3) |
| 9        | 0.9370 (2)          | 0.9413 (1)    | 0.7022 (3) |
| 10       | 1.5676 (3)          | 0.4726 (2)    | 0.2886 (1) |
| Avg Rank | 3.0                 | 2.0           | 1.0        |

The Friedman test showed a significant difference in accuracies ( $\chi_F^2 = 20.0 > 7.8$ ,  $p < 0.05$ ), indicating that the algorithms' accuracy results are statistically different. The Nemenyi test revealed a significant difference between Random Forest vs. SVM (diff = 2.0 > CD = 1.0478). No significant differences were observed between Logistic Regression and Random Forest (diff = 1.0 < CD) and Logistic Regression and SVM (diff = 1.0 < CD).

#### C. F1-Score

TABLE III. F1-SCORE RESULTS

| Data set | Logistic Regression | Random Forest | SVM        |
|----------|---------------------|---------------|------------|
| 1        | 0.8895 (2)          | 0.9389 (1)    | 0.5529 (3) |
| 2        | 0.9020 (2)          | 0.9441 (1)    | 0.5597 (3) |
| 3        | 0.9132 (2)          | 0.9500 (1)    | 0.5612 (3) |
| 4        | 0.9188 (2)          | 0.9580 (1)    | 0.5723 (3) |
| 5        | 0.8901 (2)          | 0.9311 (1)    | 0.5657 (3) |
| 6        | 0.9227 (2)          | 0.9441 (1)    | 0.5652 (3) |
| 7        | 0.9157 (2)          | 0.9474 (1)    | 0.5466 (3) |
| 8        | 0.9096 (2)          | 0.9438 (1)    | 0.5659 (3) |
| 9        | 0.9155 (2)          | 0.9235 (1)    | 0.5387 (3) |
| 10       | 0.8901 (2)          | 0.9375 (1)    | 0.5455 (3) |
| Avg Rank | 2.0                 | 1.0           | 3.0        |

The Friedman test demonstrated a significant difference in F1-scores ( $\chi_F^2 = 20.0 > 7.8$ ,  $p < 0.05$ ), confirming that the algorithms differ significantly in terms of F1-scores. The Nemenyi test revealed a significant difference between Random Forest vs. SVM (diff = 2.0 > CD = 1.0478). No significant differences were observed between Logistic Regression and Random Forest (diff = 1.0 < CD) and Logistic Regression and SVM (diff = 1.0 < CD).

#### D. Discussion and Conclusion

The results indicate significant differences among the classifiers in all three evaluation measures. The Nemenyi test reveals that:

- SVM is significantly faster in training time compared to Logistic Regression.
- Random Forest significantly outperforms SVM in accuracy and F1-score.
- No significant differences are observed between Logistic Regression and Random Forest in any measure.

Random Forest consistently performs well across all metrics, making it the most balanced choice for this dataset.

#### REFERENCES

- [1] [1] H. Witten, E. Frank, M. Hall, and C. Pal, *Data Mining: Practical Machine Learning Tools and Techniques*, 4th ed. Morgan Kaufmann, 2016
- [2] [2] M. Friedman, *The Use of Ranks to Avoid the Assumption of Normality Implicit in the Analysis of Variance*, Journal of the American Statistical Association, vol. 32, no. 200, pp. 675–701, 1937.
- [3] [3] P. Nemenyi, *Distribution-Free Multiple Comparisons*, Ph.D. dissertation, Princeton University, 1963.
- [4] [4] UCI Machine Learning Repository, *Spambase Dataset*. [Online]. Available: <https://archive.ics.uci.edu/ml/datasets/spambase>
- [5] [7] J. Demšar, Statistical Comparisons of Classifiers over Multiple Data Sets, *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.